



Data for finance: databases, SQL, DSA, KDB+/Q

How to efficiently store, query and structure financial data, including at high frequency.

ALGORITHMIC COMPLEXITY

$O(1)$ direct access < $O(\log n)$ sorted search < $O(n)$ simple scan < $O(n^2)$ pairwise comparison

RELATIONAL DATABASES

- Organize data into tables linked by keys (primary/foreign key).
- Guarantee data integrity through constraints.
- The standard for transactional data (positions, orders, counterparties).

SQL

- Universal language for querying and manipulating relational databases.
- SELECT / JOIN / GROUP BY: the 3 building blocks to master first.
- A skill expected almost everywhere in quantitative finance.

DSA (DATA STRUCTURES)

- Lists, trees, hashmaps, queues: the choice of structure directly impacts performance.
- An order book is often modeled with trees or sorted queues.
- Essential for any low-latency trading system.

KDB+/Q

- Columnar database specialized in very high-frequency time series.
- Used by high-frequency trading desks to store and query tick data.
- The Q language is notoriously hard but extremely fast for this use case.

Who uses what?

DATA ENGINEER

Designs and maintains the organization's pipelines and databases.

QUANT DEVELOPER

Uses DSA to optimize the performance of pricing/risk engines.

ANALYST

Queries data via SQL to produce analysis and reporting.

HIGH-FREQUENCY TRADER

Relies on KDB+/Q to analyze tick data in near real time.

KEY TAKEAWAYS

1. SQL remains the single most universally useful data skill in finance, ahead of any specific programming language.
2. The choice of data structure (DSA) becomes critical as soon as latency or data volume grows significantly.
3. KDB+/Q only makes sense for very large volumes of tick-by-tick data — overkill for a monthly report.

Comparison table

CRITERION	SQL (RELATIONAL)	KDB+/Q	NOSQL
Data structure	Linked tables	Columns, time series	Flexible documents/key-value
Typical use case	Positions, transactions	High-frequency tick data	Unstructured data, logs
Time-series performance	Decent	Excellent	Variable

Real-world examples

SQL QUERY

Extract a security's 5-year price history with a simple SELECT ... WHERE ... ORDER BY date.

KDB TABLE FOR TICK DATA

Store millions of trades a day and compute a VWAP in milliseconds thanks to KDB+'s columnar storage.

HASHMAP FOR LOOKUPS

Map an instrument ID to its attributes with instant $O(1)$ access instead of scanning a list.

TREE FOR AN ORDER BOOK

An order book is often implemented with a balanced tree to quickly find the best bid/ask price.

Quick glossary

PRIMARY KEY

Unique identifier of a row in a database table.

INDEX

Structure that speeds up search within a database.

Q LANGUAGE

Query language associated with KDB+, geared toward time-series processing.

TIME-SERIES DATABASE

Database optimized for storing and querying time-stamped data.

BIG-O

Notation that measures an algorithm's complexity/performance as data size grows.

ORDER BOOK

Real-time record of buy and sell orders for an instrument.

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